



Kaleido

EXECUTIVE OVERVIEW

Typed. Discoverable. Stateful.

A .NET framework for exposing typed queries and stateful business processes as machine-discoverable HTTP capabilities.



Workflows • channels • interoperability • AI
OPEN-SOURCE PROJECT OVERVIEW

PRE-1.0 PROJECT PREVIEW

THE IDEA AT A GLANCE

One contract layer for
humans, applications,
and AI.

01

Discover

What capabilities
exist?

02

Invoke

How do I call them?

03

Continue

What comes next?

**One business
capability model.
Many consumers.**



Kaleido

Typed. Discoverable. Stateful.

A .NET framework for building discoverable business
queries and stateful processes over HTTP.

.NET



Open Source



MIT License



AI Friendly



Discoverable

Self-describing APIs
with rich metadata.



Stateful Processes

Long-running, multi-step
workflows with a clear
process model.



Typed Queries

Expose data through
strongly typed, queryable
endpoints.



Extensible

Clean extension points
where they add real value
(and no more).



Built for AI

Machine-readable
capabilities designed
for modern AI agents
and tools.

Kaleido helps you expose your business capabilities as
machine-discoverable, typed, and stateful APIs. It's designed for
developers building modern applications, integrations, and AI agents.



Get Started →



Documentation →

```
C# HTTP JSON
// Execute a discovered process step
var client = new KaleidoClient(new Uri("https://api.example.com"));
var response = await client.Processes
    .For("orders")
    .StartAsync(new { customerId = 123 });
1
Console.WriteLine($"Process started: {response.ProcessId}");
```

Three primitives make the system understandable

Kaleido turns ordinary .NET types into consistent, discoverable runtime contracts.



Queryable

What does the business know?

- Typed views and parameters
- Search, filter, sort, and page
- Published field and constraint metadata

INFORMATION CAPABILITIES

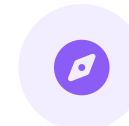


Process

What can the business do?

- Stateful, multi-step actions
- Dependencies and availability rules
- Next-step guidance through process state

ACTION CAPABILITIES



Registry

How does a consumer find it?

- Capability catalog and metadata
- Advertised execute and query URLs
- Local and downstream aggregation

DISCOVERY CAPABILITIES

ORDINARY .NET TYPES → VALIDATED METADATA → CONSISTENT HTTP CAPABILITIES

Correlation • clients • observability • persistence

A shared capability layer—without replacing your platforms

Define business capabilities once, then let workflow tools, channels, partners, and agents consume the same contracts.



Kaleido complements orchestration platforms, channel runtimes, and interoperability standards.

The value is the shared contract between them.

What an AI agent can do with Kaleido

Kaleido gives an agent a map of business capabilities—not unrestricted authority.



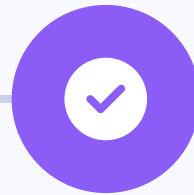
Discover

Read the registry



Understand

Inputs, outputs, constraints



Choose

Select a valid capability



Execute

Call the advertised URL



Continue

Use processId + next steps

ALREADY AVAILABLE

Machine-readable operating context

- Capability names, descriptions, schemas, and constraints
- Advertised query and execution URLs
- Explicit process state and next-step guidance
- Structured results, errors, and correlation context

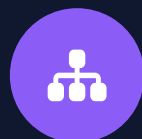
BEFORE AUTONOMOUS ACTION

Govern the decision boundary

- Authorization-aware discovery and allowlists
- Side-effect, confirmation, and data-sensitivity metadata
- Durable idempotency and concurrency guarantees
- Registry freshness, redaction, and resource limits

Initial target consumers

Kaleido is the reusable capability layer beneath the systems that orchestrate work, interact with people, or exchange standards-based data.



WORKFLOW ENGINES

Pega and similar platforms

Keep case orchestration, SLAs, assignments, and human work in the workflow platform. Call Kaleido for reusable, typed business capabilities.

FEWER BESPOKE CONNECTORS

SHARED BUSINESS CONTRACTS

Kaleido complements case management; it does not replace it.



IVR / CONTACT CENTER

Conversational and assisted channels

Use metadata to drive prompts and validation, preserve an explicit process ID, and resume work across transfers, agents, or digital handoffs.

CONSISTENT PROMPTS

CROSS-CHANNEL CONTINUITY

The same backend action is available to voice, agent desktop, and web.



EXTERNAL INTEGRATIONS

Da Vinci CRD · DTR · PAS

Handle FHIR- and X12-oriented interactions at the edge, then map them into stable internal queries and process steps that can be reused elsewhere.

STANDARDS AT THE EDGE

STABLE INTERNAL MODEL

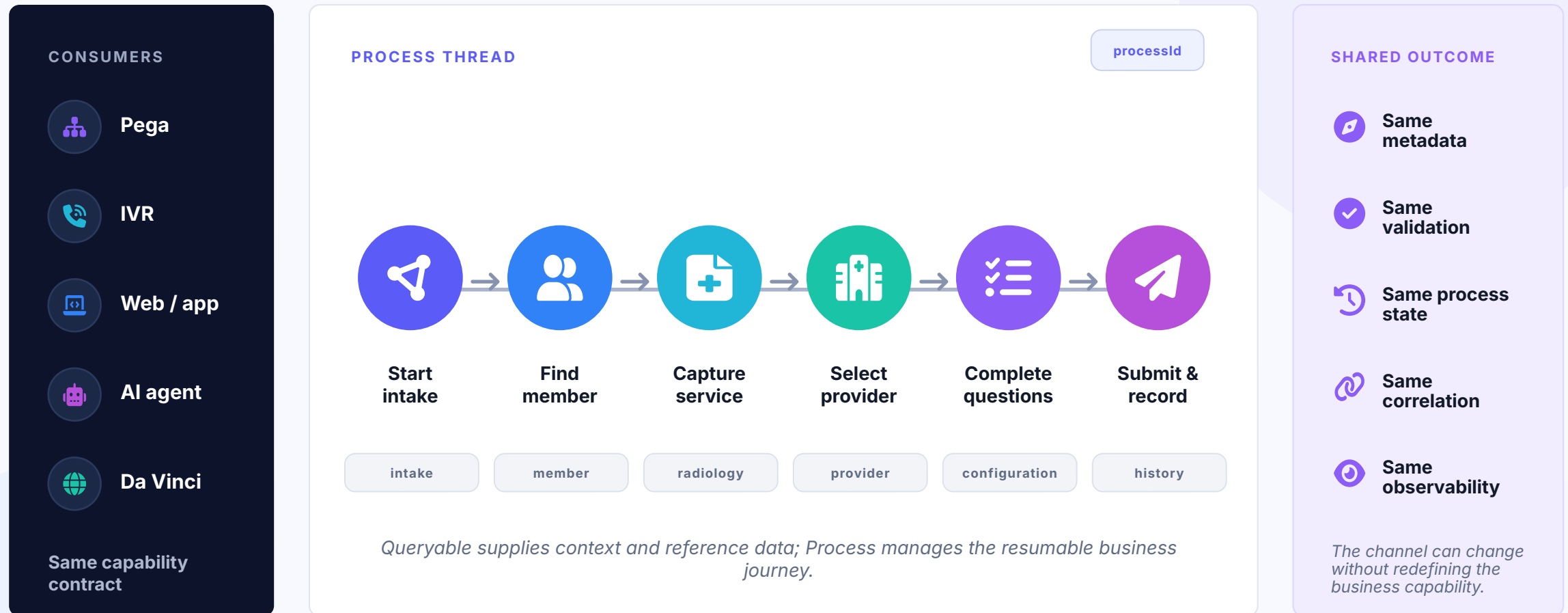
CRD discovers requirements, DTR gathers documentation, and PAS supports submission.

Complementary, not replacement.

Sources: [Pega workflow automation](#) • [HL7 Da Vinci CRD 2.2.1](#), [DTR 2.2.0](#), [PAS 2.2.1](#)

One prior authorization process, many consumers

A single process can span multiple services while every channel uses the same metadata, validation, and state model.



EXECUTIVE TAKEAWAY

Standardize once. Reuse everywhere.

Kaleido creates a common language between business services and the systems that need to use them.



Shared capability contracts

A consistent way to expose queries and actions.



Less one-off integration

Channels reuse metadata, URLs, validation, and state.



Resumable, observable work

Explicit process identity and correlation across services.



Governed path to AI

Supervised tool use now; safer autonomy as controls mature.

NEAR-TERM DIRECTION



01 PREVIEW

Publish and validate consumer packages



02 HARDEN

State, idempotency, privacy, authorization



03 PROVE

Pega, IVR, and Da Vinci adapters



04 SCALE

Architecture and tech-lead guidance